KATIE C. W. DAMES, Ph.D.

Applied Wetland Ecologist | Natural Resources Project Scientist | Environmental Scientist

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PROFESSIONAL SUMMARY

Doctoral-trained Applied Wetland Ecologist and Environmental Scientist with 7+ years of field ecology, aquatic/wetland ecosystem assessment, environmental sampling, laboratory analysis, spatial analysis, and technical reporting experience across freshwater, wetland, marine, forested, and estuarine systems. Specializes in translating field and laboratory data into defensible conclusions for habitat condition, ecosystem stress, restoration, mitigation, and environmental decision-making. Experienced planning field campaigns, mentoring junior researchers, coordinating interdisciplinary teams, producing proposal materials and technical reports, and communicating complex science to faculty, agency, stakeholder, and community audiences.

CORE CONSULTING COMPETENCIES

Wetland and aquatic ecosystem assessment | Ecological and natural resource field surveys | Habitat condition and impact assessment | Wetland restoration and mitigation frameworks | Soil, water, eDNA and biodiversity sampling | Field/lab data QA/QC and defensible interpretation | Technical reports, proposals and research briefs | Environmental protection and monitoring plans | WOTUS, Section 401/404 and permitting concepts | Agency, stakeholder and interdisciplinary coordination | Project coordination and junior staff mentoring | Safety, quality and professional field standards

PROFESSIONAL EXPERIENCE

Doctoral Research Fellow / Applied Wetland Ecologist | Tuskegee University | *Tuskegee, AL* | Aug 2022 - May 2026

- Led applied wetland ecology research assessing climate-driven ecological simplification of forested vernal pools using bioindicators, soil/water chemistry, environmental DNA, remote sensing, and climate data.
- Planned and executed multi-site field campaigns from study design through ecological surveys, environmental sampling, laboratory analysis, data interpretation, and reporting.
- Conducted wetland soil and water sampling, amphibian bioindicator assessment, microbial/eDNA workflow development, and hydroperiod/biogeochemical analysis to evaluate habitat condition and ecosystem stress.
- Integrated field observations, laboratory results, LiDAR-derived microtopography, GIS/remote sensing products, and CESM2 climate data to support restoration, monitoring, and environmental decision-making.
- Developed technical reports, proposal materials, scientific presentations, data summaries, and interdisciplinary briefings translating complex field/lab data into defensible conclusions.
- Trained and mentored undergraduate and graduate researchers in wetland field methods, molecular workflows, data management, safety expectations, and quality standards.

Graduate Visitor Program Fellow | NSF National Center for Atmospheric Research | *Boulder, CO* | Jun 2025 - Aug 2025

- Analyzed more than 73 million CESM2 Large Ensemble climate data points to quantify long-term seasonal precipitation and temperature variability relevant to wetland ecosystem stability.
- Processed multi-ensemble NetCDF outputs using R in a Linux-based high-performance computing environment; computed rolling means, variability metrics, and trend slopes.
- Collaborated with Research Applications Laboratory and Climate and Global Dynamics scientists to connect climate model outputs with ecological assessment and resilience frameworks.

Master's Graduate Research Assistant / Environmental Scientist | Tuskegee University | *Tuskegee, AL* | Aug 2020 - May 2022

- Conducted estuarine field surveys with Dauphin Island Sea Lab to evaluate anthropogenic influences on Mobile Bay ecosystem dynamics and species habitat use.
- Compiled and structured 600+ field observation records into reproducible datasets; applied spatial and statistical analyses to assess distribution patterns, habitat partitioning, and ecological responses.
- Produced a technical thesis recognized as the department's top master's project for analytical rigor, ecological synthesis, and scientific communication.
- Supported biodiversity and forestry research initiatives, including invasive species impact experiments; served as a teaching assistant for ecology and environmental science courses.

Agriculture and Environmental Science Enrichment Counselor | Tuskegee University | *Tuskegee, AL* | Jun 2022 - Jul 2022

- Supervised and mentored 30+ students in environmental, forestry, conservation, and land stewardship programming.
- Designed and led hands-on research poster projects integrating field sampling, laboratory analysis, ecosystem dynamics, and scientific communication.
- Coordinated outreach with Alabama Game Warden Service, U.S. Forest Service, and U.S. Fish and Wildlife Service representatives.

ADDITIONAL FIELD ECOLOGY AND RESTORATION EXPERIENCE

Undergraduate Research Assistant - Marine and Coastal Ecology | University of South Florida Study Abroad | *Soufriere, Saint Lucia* | Sep 2017 - May 2018

- Conducted quantitative benthic and coral reef ecosystem assessments; completed 25+ scientific SCUBA surveys measuring reef structure, fish biomass, and habitat composition.
- Analyzed and presented ecological survey data to Soufriere Marine Management stakeholders; participated in coral reef and coastal restoration activities within a marine protected area.

Marine Ecology Research Student | Duke University Marine Lab | *Beaufort, NC* | May 2017 - Jul 2017

- Designed and conducted field studies quantifying marine invertebrate communities across beach face, dock, estuarine, and rocky intertidal habitats.
- Developed and presented a seagrass restoration proposal emphasizing habitat recovery, ecosystem resilience, and applied coastal management.

Freshwater Nutrient Cycling Research Student | Chicago Shedd Aquarium Research Lab | *Chicago, IL* | Dec 2014 - 2016

- Designed and executed freshwater nutrient cycling experiments focused on nitrogen and phosphorus dynamics, non-point source pollution, and Lake Michigan eutrophication.
- Collected and analyzed water chemistry data to evaluate nutrient-driven ecosystem responses and presented findings to research mentors and peers.

TECHNICAL SKILLS AND CERTIFICATIONS

Wetland / Environmental Methods: soil and water chemistry sampling; eDNA extraction and QC; ecological field surveys; habitat assessment; biodiversity and bioindicator monitoring; field data QA/QC; environmental sampling protocols

Data / Modeling: R, Python, SAS, IBM SPSS, PRIMER, CESM2 Large Ensemble, NetCDF workflows, Linux/HPC environments, multivariate statistics, trend and variability analysis

Spatial / Remote Sensing: ArcGIS Pro, QGIS, ERDAS Imagine, LiDAR-derived elevation models, remote sensing workflows, spatial analysis, kernel density interpolation

Certifications / Field Safety: Esri Basic GIS Certification; PADI Open Water Diver; AAUS Scientific Diver; CPR and marine field safety training

EDUCATION

Ph.D., Integrative Biosciences - Tuskegee University, Tuskegee, AL | 2026

M.S., Environmental Science - Tuskegee University, Tuskegee, AL | 2022

B.S., Marine Biology - University of South Florida, Tampa, FL | 2020

Dissertation: Assessing the Effects of Climate-Driven Ecological Simplification of Vernal Pools in Tuskegee Forest Systems Using Bioindicators

SELECTED HONORS, GRANTS AND PRESENTATIONS

- NSF NCAR Graduate Visitor Program Fellowship Recipient (2025).
- USDA Future Leaders in Agriculture Recipient (2024).
- Co-author, 1890 Universities Foundation USDA Urban, Community Forestry and Climate Response $5M Grant (2023).
- Agriculture and Environmental Sciences Department Top Master's Thesis Honor (2022).
- Presented climate variability and ecological simplification research at Tuskegee CAENS Seminar and NSF NCAR/UCAR/UCP Summer Student Conference (2025).

COMMUNITY AND CONSERVATION LEADERSHIP

- Led regional Alabama Black Belt Counties Science Fair programming across Macon, Lowndes, and Bullock counties; developed scientific-method curriculum and mentored K-8 students in experimental design and science communication.
- Managed budget and conservation-focused fundraising for Tuskegee University Ducks Unlimited Conservation Club; supported wildlife stewardship and outreach programming.